We consider the unconstrained optimization problem

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \quad f(x), \tag{1}$$

where $f\colon \mathbb{R}^n \to \mathbb{R}$ is a continuously differentiable nonconvex function, and $x \in \mathbb{R}^n$ is the optimization variable. We focus on settings where objective function evaluations are contaminated by bounded and non-diminishing numerical noise [11]. Specifically, only inexact function values $\overline{f}(x)$ are available for optimization. Such inaccuracies are unavoidable in many realistic scenarios, including finite-precision arithmetic, simulation-based evaluations, and stochastic approximations [1, 6, 10]. When function values are contaminated by numerical noise, standard optimization algorithms may lose their theoretical guarantees, behave erratically, or terminate prematurely with unreliable solutions.

In the absence of noise and with accurate function evaluations, quasi-Newton methods, in particular the Broyden–Fletcher–Goldfarb–Shanno (BFGS) method and its Limited-memory variant (L-BFGS), are among the most efficient and widely used algorithms to solve the problem (1) [5, 7]. Their success is largely attributed to fast local convergence and scalability. Standard implementations [8, 13] rely on line search procedures that enforce the Wolfe conditions to guarantee sufficient descent and the positive definiteness of the Hessian approximation.

In the presence of numerical noise, these line search conditions may become unreliable. Differences in function values or directional derivatives can be dominated by rounding errors, leading to unstable step sizes, ill-conditioned Hessian approximations, or abnormal termination [9, 10]. From a computational perspective, improving the robustness of quasi-Newton methods in noisy environments is therefore crucial. In particular, it is desirable to design algorithms that (i) retain computational efficiency when function evaluations are accurate, (ii) remain stable when function values are unreliable, and (iii) guarantee convergence to first-order stationary points under mild assumptions.

In this work, we propose a noise-tolerant regularized quasi-Newton method that addresses these challenges. Our approach combines Armijo-type line search with quadratic regularization [4, 12] and Objective-Function-Free Optimization (OFFO) inspired strategy [2, 3]. When a sufficient decrease in function value is observed, the method behaves similarly to standard line-search-based quasi-Newton methods, preserving its practical efficiency. When numerical noise dominates, we introduce regularization on the approximate Hessian to ensure numerical stability, which yields convergence properties similar to those of OFFO methods, particularly AdaGrad-Norm [14]. The resulting algorithm is particularly well-suited for low-precision or noisy environments.

From a theoretical standpoint, we establish that the proposed method achieves the standard global convergence rate of $\mathscr{O}\left(1/\varepsilon^2\right)$ for smooth nonconvex optimization problems, even when objective function values are corrupted by noise. Extensive numerical experiments on the CUTEst benchmark collection, across various floating-point precisions and artificial noise settings, demonstrate that the proposed method is significantly more stable than standard line-search-based methods in settings with significant numerical noise, while maintaining practical convergence speed.

References

1. Caflisch RE (1998) Monte Carlo and quasi-Monte Carlo methods. Acta Numerica 7:1–49, DOI 10.1017/S0962492900002804
2. Gratton S, Toint PL (2023) OFFO minimization algorithms for second-order optimality and their complexity. Computational Optimization and Applications 84(2):573–607, DOI 10.1007/s10589-022-00435-2
3. Gratton S, Jerad S, Toint PhL (2024) Complexity of a class of first-order objective-function-free optimization algorithms. Optimization Methods and Software 0(0):1–31, DOI 10.1080/10556788.2023.2296431
4. Kanzow C, Steck D (2023) Regularization of limited memory quasi-Newton methods for large-scale nonconvex minimization. Mathematical Programming Computation 15(3):417–444, DOI 10.1007/s12532-023-00238-4
5. Liu DC, Nocedal J (1989) On the limited memory BFGS method for large scale optimization. Mathematical Programming 45(1):503–528, DOI 10.1007/BF01589116
6. Moré JJ, Wild SM (2011) Estimating Computational Noise. SIAM Journal on Scientific Computing 33(3):1292–1314, DOI 10.1137/100786125
7. Nocedal J, Wright SJ (1999) Numerical Optimization. Springer, DOI 10.1007/978-0-387-40065-5
8. Okazaki N (2007) libLBFGS: A library of limited-memory broyden-fletcher-goldfarb-shanno (l-BFGS)
9. Paquette C, Scheinberg K (2020) A Stochastic Line Search Method with Expected Complexity Analysis. SIAM Journal on Optimization 30(1):349–376, DOI 10.1137/18M1216250
10. Shi HJM, Xie Y, Byrd R, Nocedal J (2022) A Noise-Tolerant Quasi-Newton Algorithm for Unconstrained Optimization. SIAM Journal on Optimization 32(1):29–55, DOI 10.1137/20M1373190
11. Sun S, Nocedal J (2023) A trust region method for noisy unconstrained optimization. Mathematical Programming 202(1):445–472, DOI 10.1007/s10107-023-01941-9
12. Ueda K, Yamashita N (2014) A regularized Newton method without line search for unconstrained optimization. Computational Optimization and Applications 59(1):321–351, DOI 10.1007/s10589-014-9656-x
13. Virtanen P, Gommers R, Oliphant TE, Haberland M, Reddy T, Cournapeau D, Burovski E, Peterson P, Weckesser W, Bright J, van der Walt SJ, Brett M, Wilson J, Millman KJ, Mayorov N, Nelson ARJ, Jones E, Kern R, Larson E, Carey CJ, Polat I, Feng Y, Moore EW, VanderPlas J, Laxalde D, Perktold J, Cimrman R, Henriksen I, Quintero EA, Harris CR, Archibald AM, Ribeiro AH, Pedregosa F, van Mulbregt P, SciPy 10 Contributors (2020) SciPy 1.0: Fundamental algorithms for scientific computing in python. Nature Methods 17:261–272, DOI 10.1038/s41592-019-0686-2
14. Ward R, Wu X, Bottou L (2020) AdaGrad stepsizes: Sharp convergence over nonconvex landscapes. J Mach Learn Res 21(1):219:9047–219:9076